

TASK 4.1-P1: TRANSFORMERS

Name: Zulekha D/O Muhammad Usman

- What transformers solved:

The previous models like ANN were each limited either by certain function, memory or data types, although they're not completely useless they are still used on small data unlike the transformer that is built to use on more complicated data, and memorize it for a longer term since many of those models lacked a proper memory and started forgetting the start of the sentence till they reach the end of it.

for example ANN worked only on fixed size data, while CNN works only on image classification and recognition so it struggles with text, as for RNN they were used to solve the issue of ANN, but at certain point it flared up with “**vanishing gradient descent problem**”, even with LSTM it can't handle as much, on the contrary of those transformers started for NLP but now is used in many different fields including computer vision that too while keeping up with all the words from the start and performing tasks in parallelism.

- Attention:

Attention means transformers started connecting and creating context for words like we humans do, for example if i said “this is my brother, his name is Mustafa” it will create an attention to relate “Mustafa” and “brother” (also keep in mind it cares about sequence in masked self attention), so “brother” wouldn't have access to “Mustafa”, but vice versa is possible, as shown:

	<start>	I	am	fine
<start>	1	0	0	0
I	0.37	0.62	0	0
am	0.26	0.31	0.43	0
fine	0.21	0.26	0.26	0.26

where every word creates an attention weight with itself and the previous words only

- The context window:

Unlike it seems, transformer isn't a perfect model, since transformers access large data at once where each word calculates a relationship with every other one, it creates a limit to how much the transformer can read or write words, eventually when it reaches that limit it drops the extra data off, and because of the mathematical formulas used to create those models it ends up with “Quadratic complexity” where it explodes/stops working due to high computational power